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ASSINGMENT

ON

AI ETHICS

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Section :11

Case Study Assigned: Chapter 4 (Case 1: Data Appropriation)

Surveillance Capitalism (Case 1)

CO4 — What AI application is it? Impact & sustainability

Privacy leakage is defined as the extent to which an attacker can learn about a person's generative biometric data based on reference data stored of the individual.

Commercial facial recognition is constructed by scraping (scraping) user photos in bulk at social and web sites; then scraped information is returned to train/run identification and search services marketed to third parties. The fact that these systems gathered billions of scraped photos (e.g, Clearview AI) has been widely reported in various technical literature.

Xinai Electronics has collected millions of license plates and facial images, by camera networks which it claimed to have kept hidden. However, since the information was not stored in a safe way, anyone with the appropriate information could access it with the help of a web browser, which presents legitimate concerns regarding the breach of privacy. **(Laishram et al., 2024)**

From a societal point of view it Causes serious privacy and autonomy threats (individuals did not agree to downstream biometric uses), reinforces power disparities between data corporations and the citizens, and may facilitate invasive surveillance in unexpected locations. It is socially unsustainable to make non-consensual biometric surveillance since it gradually undermines trust. It threatens the privacy and civil liberties greatly, despite getting more accurate. There is a high possibility of weaponization of such technologies and their misuse. Hence there is a need to enhance and institute more holistic policies, corporation and ethical principles to ensure vulnerable and minority groups are not harmed as this technology keeps advancing. **(Raji & Buolamwini, n.d.)**

CO5 — Key ethical issues & how to address them (with research support)

The key ethical issues would be

Privacy and Consent: Pictures were taken for one reason and then used for face monitoring, which went against the principles of consent and purpose

Fairness/bias: FR performance might differ by gender and skin tone, which increases the possibility of biased results

Accountability and transparency: Training data cannot be viewed, disputed, or removed; as data sources and models are opaque.

The way these ethical challenges can be addressed

Require white-/inside-the-box audit access via procurement.

Facial recognition systems that you cannot explore fully should not be purchased. We need to insist on the data, testing tools, and documentation to guarantee that we are at a position to conduct strict audits. Auditability is another contract requirement that should be made when purchasing. (Casper et al., AIES 2024; Johnson et al., FAccT 2025)

Auditing and disclosure of training data and error rate is necessary to promote more fairness and accountability to the vendors. This must be supplemented by strong data governance not just in the sense of transparent data origin, user opt-out and impact measurement, but in deployment as well.

CO6 — Social/legal concerns & the responsibilities of engineers (with research support)

The post-deployment concerns would be

Primary issues with facial recognition technology are intrusion of privacy and snooping by unwarranted monitoring. Function and mission creep is an issue in which its application. often goes beyond its original purpose, and threatens unexpectedly. Lastly, the system may be operationally vulnerable with mediocre performance in the practical environments and vulgar to spoofing.

Engineers' responsibilities before deployment

Enhancing Transparency: Engineers are required to publish official documents including Model Cards which have the ability to clarify the purpose of the system to be implemented, the data they are designed on, how they will work in various groups of people as well as the areas that are known to be weak. (Mitchell et al., 2019)

Supply Chain Management : It should be noted to map out all the suppliers and services that were used to develop the system. This aids in the distribution of blame during instances of failures and in making all the technical facts audit-able.

Installing post-deployment monitoring with a Coordinated Flaw Disclosure (CFD) : channel such that researchers and users report harms or failures and we can verify, triage and remediate--to and including kill-switches where necessary. (Cattell, Ghosh, & Kaffee, 2024).

Conducting Regular Audits: Frequent audits should be done like the use of the third-party and error-checking in order to determine the existence of harm. Should an audit show that the technology is inflicting enormous harm, it should not be used any longer.

References

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